

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms, fine-tuning domain LLMs (Qwen2.5-7B, E5-large-v2) on A100 GPUs, and building automated evaluation control planes. Specialized in Python, high-throughput LLM serving internals (vLLM contributor), workflow orchestration (SAP Joule 2.0, LangGraph), and cloud-native infrastructure (Azure, AWS, Docker, Kubernetes).

Skills

- Languages & Backend:** Python, SQL, FastAPI, PostgreSQL, Oracle, REST APIs, WebSockets, AsyncIO, Linux
- AI & ML Engineering:** Agentic AI, Model Context Protocol (MCP), Multi-Agent Systems, Tool RAG, Healthcare NLP, vLLM Serving
- AI & Workflow Frameworks:** LangChain, LangGraph, vLLM, PyTorch, Hugging Face, SAP Joule 2.0, vLLM Tool Parsers, LlamaIndex, GROBID
- Cloud & Infrastructure:** Azure, AWS, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, Git, Gerrit), Redis, PGVector
- Testing, QA & Security:** PyTest, LLM-as-a-Judge, Prompt Lab, Red Teaming, PII Sanitization, Unit/Integration/Regression Testing

Work Experience

- Bigscal Technologies Pvt. Ltd.**
AI/ML Engineer Oct 2024 - Present
Surat, India
- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants on cloud infrastructure (Azure, AWS).
 - Architected multi-agent routing with **LangGraph & SAP Joule 2.0** across 650+ dynamic MCP capabilities, engineering a capability contract layer in a single ReAct loop that reduced latency by 3x and token costs by 70%.
 - Fine-tuned and served **Qwen2.5-7B & E5-large-v2** models on A100 GPUs using PyTorch, Unsloth, and contrastive learning, deploying via vLLM on Azure for structured role mapping and semantic retrieval.
 - Implemented a 4-level hierarchical RAG pipeline in Python indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a **90%+ Recall@10** retrieval rate.
 - Developed the Prompt Lab evaluation control plane in Python to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.
- Bigscal Technologies Pvt. Ltd.**
AI/ML Engineer Intern Apr 2024 - Sep 2024
Surat, India
- Engineered clinical summarization pipelines in Python for 116K+ medical records with 4-bit quantized Llama 3, and built real-time WebSocket streaming for Indic ASR with **NeMo RNN-T & CTC decoding** (STFT, Mel-spectrograms, CNN architectures).

Projects

- vLLM High-Performance Inference Engine (vllm-project/vllm)** | Open Source
Merged **PR #55475** fixing structural XML tool parser drops on ``/v1/chat/completions`` and Pydantic parameter indexing (64/64 Qwen tests passed); engineered ``torchcodec`` multimodal decoder guards (Issue #54097).
Python, PyTorch, Pydantic, AsyncIO, PyTest
- Job Architecture Generator (Qwen2.5-7B Domain SFT & Serving)** | LLM Training
Fine-tuned **Qwen2.5-7B** on 10K+ domain examples using Unsloth & PyTorch on NVIDIA A100s for role standardization and structured JSON generation; deployed on Azure via **vLLM** for low-latency enterprise inference.
Python, PyTorch, Hugging Face, Unsloth, vLLM, Azure A100
- Job Framework Semantic Transformer (e5-finetuned-job-framework)** | Model Training
Fine-tuned an ~335M-parameter **E5-large-v2** transformer on Lightcast labor data using triplet contrastive learning with hard-negative mining in PyTorch, replacing general embeddings to optimize skill retrieval precision.
Python, PyTorch, Hugging Face, E5-large-v2
- Modular Agent Harness** | Project
Principal Creator & Architect of a modular Python agentic framework implementing a formal ReAct loop with step budgets, pgvector tool routing, and dynamic MCP settings caching.
Python, FastAPI, PostgreSQL, WebSockets, pgvector

Education

- Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University** 2022 - 2024

Certifications

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| Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft |
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